

HAT-P-3b

R-1423

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-3_160602 · created 2026-08-01T06:30:57+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457541.8555 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.122 +/- 0.01
Transit depth (Rp/Rs)^2	1.48 +/- 0.26 [%]
Orbital Inclination (inc)	89.58 +/- 1.04
Airmass coefficient 1 (a1)	1.053 +/- 0.012
Airmass coefficient 2 (a2)	-0.0286 +/- 0.0017
Scatter in the residuals of the lightcurve fit is	1.16 %
Best Comparison Star	#3 - [397, 213]
Optimal Aperture	3.89
Optimal Annulus	12.99
Transit Duration (day)	0.0982 +/- 0.0024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.122 $\pm$ 0.01 vs 0.1063 (deviation 1.15 σ)
Duration fit vs published	0.0982 d vs 0.0867 d (deviation 3.51 σ)
Mid-time O-C	-5.5 min
Depth SNR	8.9
Residual scatter	1.16 %

Light curve

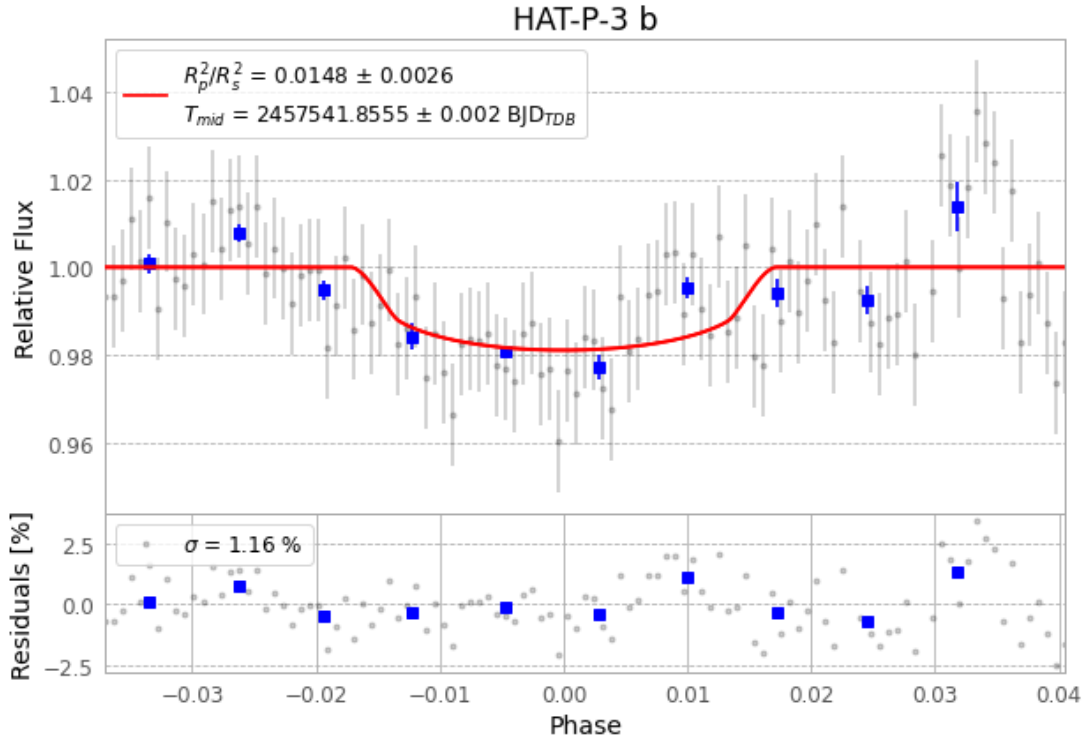


Figure 1 — FinalLightCurve_HAT-P-3 b_2016-06-01.png (SHA-256 aa41643f3bc3f5d9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [299, 232], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 cf060ccac8141228...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

109 FITS frames (72 MB), HATP-3160602055412.FITS → HATP-3160602111812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-3-160602.log (6843c92c3e94...), FinalLightCurve_HAT-P-3 b_2016-06-01.png (aa41643f3bc3...), FinalLightCurve_HAT-P-3 b_2016-06-01.csv (0740270c7f0a...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 06:30Z.